Closed-Loop Voltage and Current Control of a Buck Converter Using Cascaded PI Controllers

1. Objective

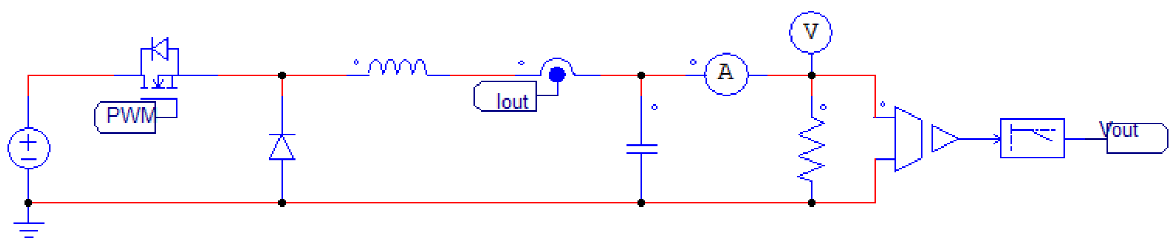
This study examines a cascaded control structure for a DC-DC buck converter using an inner current loop and an outer voltage loop. The control system uses PI controllers and is evaluated through time-domain simulation.

The objective is to demonstrate how cascaded control improves voltage regulation, current control, and overall system stability.

2. System Configuration

The system consists of:

- A PWM-controlled buck converter power stage
- An inductor current sensing path
- An output voltage sensing path
- Cascaded PI control loops:
    - Outer voltage control loop
    - Inner current control loop



The voltage loop generates a current reference, while the current loop directly controls the PWM duty ratio.

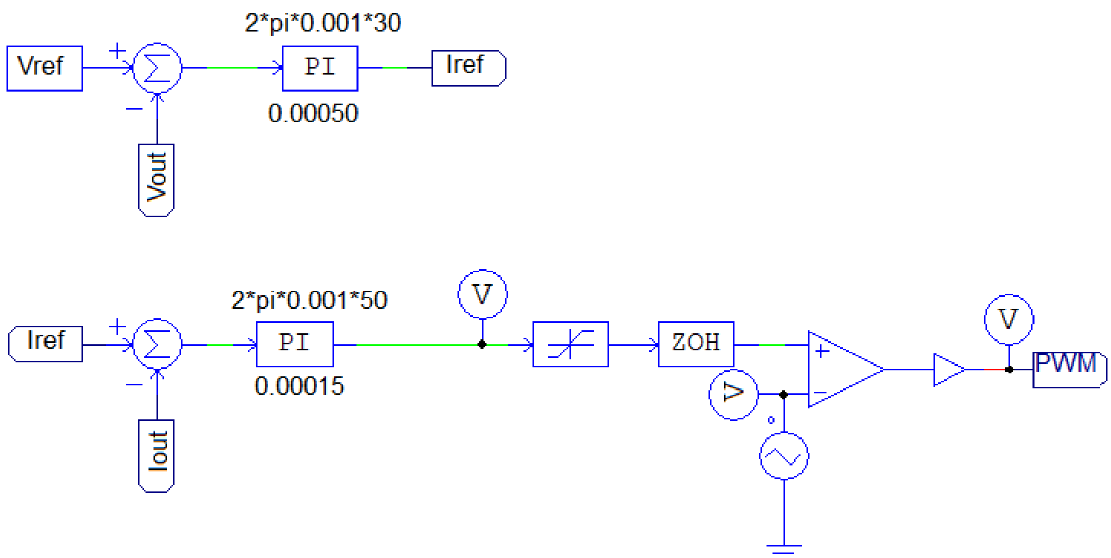
3. Control Architecture

3.1 Cascaded Loop Structure

- Outer loop: Voltage control
- Inner loop: Inductor/output current control

Control signal flow:

Vref → Voltage PI → Iref → Current PI → Duty → PWM → Power Stage



This structure ensures fast current dynamics while maintaining accurate voltage regulation.

The PI controller consists of a proportional term and an integral term. The integral action ensures zero steady-state error, while the zero introduced by the PI compensates the plant pole and improves closed-loop stability.

The transfer function of a PI controller is defined as

$$G(s) = K\frac{1 + sT}{sT}$$

This can be rewritten into the standard PI form:

$$G(s) = K\left(1+\frac{1}{sT}\right)$$

where

- K is the proportional gain

- T is the integral time constant

From this expression:

- Proportional term:

$$K_p = K$$

- Integral term:

$$K_i = \frac{K}{T}$$

4. Outer Voltage Control Loop

4.1 Purpose

The voltage control loop regulates the output voltage by adjusting the current reference supplied to the inner loop.

4.2 Error Definition

$e_v(t) = V_{ref} - V_{out}$

4.3 PI Controller

$G_v(s) = K_{pv} (1 + 1 / (T_{iv} s))$

- Output: Current reference (I_{ref})
- The controller bandwidth is designed lower than the current loop bandwidth to ensure loop decoupling.

5. Inner Current Control Loop

5.1 Purpose

The current loop forces the inductor/output current to track the reference generated by the voltage loop.

5.2 Error Definition

$e_i(t) = I_{ref} - I_{out}$

5.3 PI Controller

$G_i(s) = K_{pi} (1 + 1 / (T_{ii} s))$

- Output: Control voltage / modulation signal
- The controller output is limited to prevent over-modulation and ensure stable PWM operation.

6. PWM and Sampling Implementation

- The PI output is passed through:
  - Limiter (duty range protection)
  - Zero-Order Hold (ZOH) synchronized with the switching frequency
- PWM is generated by comparing the control signal with a carrier waveform.

This ensures discrete-time behavior consistent with digital control implementation.

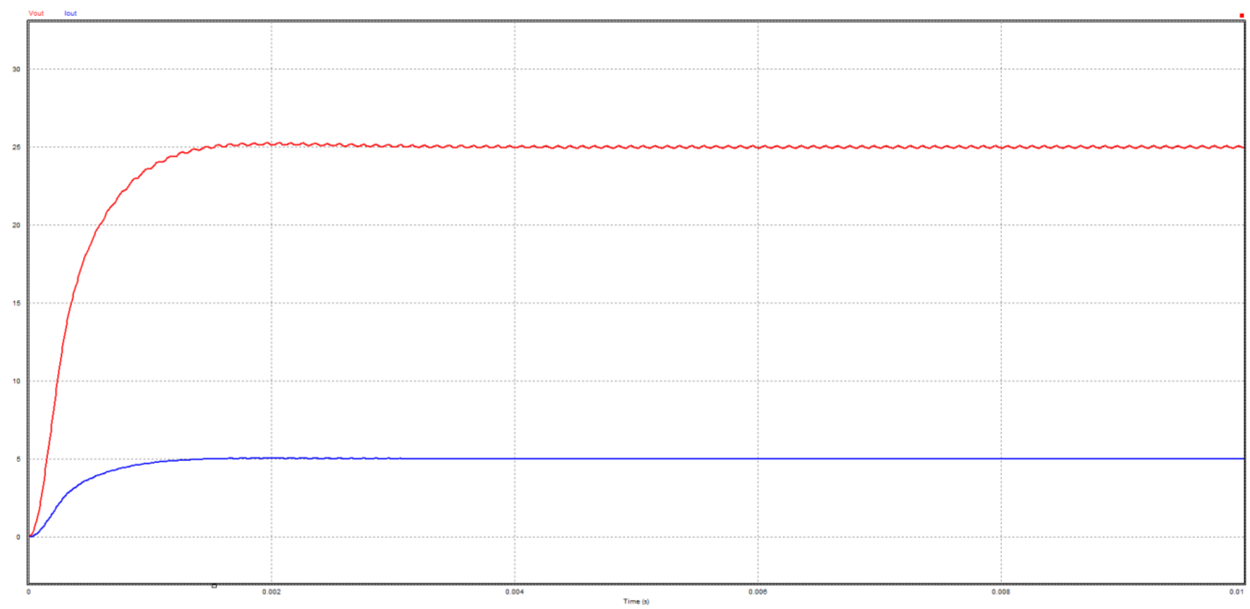
7. Design Rules

- Current loop bandwidth ≈ 5–10× voltage loop bandwidth
- Voltage loop tuned for steady regulation, not fast transient response
- Current loop tuned for fast and stable tracking
- Saturation blocks are required to avoid integrator windup and unrealistic duty commands

8. Simulation Results

Simulation confirms:

- Stable output voltage regulation
- Proper current tracking behavior
- No sustained oscillations in either loop
- Clear separation of loop dynamics



The cascaded PI structure successfully controls the buck converter under closed-loop operation.

9. Discussion

The inner current loop significantly improves system stability and transient behavior compared to single-loop voltage control.

By limiting the voltage controller output to a current reference, excessive inductor current during startup and load transients is avoided.

This cascaded architecture is widely used in practical DC-DC converters and motor drive systems.

10. Summary

A cascaded voltage–current control system using PI controllers was implemented and evaluated through simulation.

The results demonstrate stable voltage regulation, accurate current tracking, and clear separation between the inner and outer control loops.

Cascaded PI control provides a reliable and widely used control strategy for practical DC-DC power converters.